ADMINISTRATION

Exam Logistics & Overview

Test Overview

- → **Module:** Mathematics for Computing (ICT 2)
- → **Format:** Written, In-Class Test
- → **Time:** 90 Minutes
- → **Tools:** Scientific calculators allowed; Closed-book format.

Required Score

Minimum 30/100 in ICT2 specifically. Combined average with ICT1 must be at least 40%.

Main Topics

Set Theory & Logic
Graph Theory & Trees
Matrix Algebra
Statistics & Probability

SETS

Set Operations & Intervals

1. Fundamental Operations

Union ($A \cup B$): All elements in either set.
Intersect ($A \cap B$): Elements shared by both.
Difference ($A \setminus B$): In A, but not in B.
Complement (A'): Elements in Universal set NOT in A.

2. Interval Notations

$[a, b]$
Inclusive ($\leq, \geq$) - Solid dots
(a, b)
Exclusive ($<, >$) - Open dots

> **Worked Example:** Find $\{x \in A : x^2 \in A\}$ where $A = \{4, 5, \ldots, 25\}$.
> **Logic:**
> $4^2 = 16$ (Present in A ✓)
> $5^2 = 25$ (Present in A ✓)
> $6^2 = 36$ (Outside A $\times$)
> **Result:** $\{4, 5\}$.

SETS

Power Sets & Products

1. Power Set $P(A)$

The set of all possible subsets of A, including the empty set and itself.

$$|P(A)| = 2^{|A|}$$

Example: If $A = \{0, 1\}$, $P(A) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$ (Size: $2^2 = 4$).

2. Cartesian Products

$$|A \times B| = |A| \times |B|$$

The Intersection Subtraction Rule

To find the cardinality $|(A \times B) \setminus (B \times A)|$:

1. Calculate Total Pairs: $|A| \times |B|$.
2. Identify Intersection: $S = A \cap B$.
3. Calculate Shared Pairs: $|S| \times |S|$.
4. **Formula:** $(|A| \times |B|) - (|S|^2)$.

RELATIONS

Binary Relation Properties

Reflexive

$\forall a \in A, (a, a) \in R$. Every element in the set must relate to itself.

Symmetric

If $(a, b) \in R$, then $(b, a) \in R$. If there is a connection one way, it must exist the other way.

Transitive

If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$. "A shortcut" must always exist.

Equivalence Relation

A relation is an Equivalence Relation if and only if it is **Reflexive**, **Symmetric**, and **Transitive**.

Common Exam Trap: A relation like $R = \{(0, 0)\}$ on $A = \{0, 1\}$ is NOT reflexive because $(1, 1)$ is missing from the relation.

FUNCTIONS

Logic of Functions

Defining a Function

A function $f : A \to B$ requires every element in the **Domain** (A) to point to **EXACTLY ONE** element in the **Codomain** (B).

Invalid Scenarios:

- An input x points to two different outputs y_1 and y_2.
- An element in Domain A has no arrow pointing out.

Valid Scenarios:

- Two different inputs point to the same output.
- An element in the codomain has no arrows pointing to it.

> **Check:** Is $f(x) = \sqrt{x}$ a function on the domain of all Real Numbers ($\mathbb{R}$)?
> **Answer:** NO. Inputs like -1 result in no real number output, leaving parts of the domain "unmapped".

LOGIC

Truth Tables & Precedence

1. Order of Evaluation

1. $\neg$ (Not) 2. $\wedge$ (And) 3. $\vee$ (Or) 4. $\rightarrow$ (If...then)

2. Table Dimensions

To determine the size of a truth table:
Number of Rows = 2^n, where n is the count of distinct variables.

Practice: Count rows for: $\neg(p \rightarrow (q \wedge \neg q)) \vee (\neg r \rightarrow (s \vee \neg s))$

Unique Variables: $\{p, q, r, s\} \implies 2^4 = 16$ rows.

3. Logic Puzzle Strategy

When solving "Who is telling the truth?" puzzles: identify **contradictions**. If the premise says "Only one statement is true", test each scenario. The scenario that satisfies exactly one statement is the correct one.

LOGIC

Validity & Logical Consequence

Classifications

- → **Tautology:** The statement is True in every single row.
- → **Contradiction:** The statement is False in every single row.
- → **Contingency:** The result is mixed (some True, some False).

Validity Check: $((p \vee q) \wedge \neg q) \rightarrow p$
Constructing a table would show this is a **TAUTOLOGY** (all True). This is the "Disjunctive Syllogism" rule.

The Implication Trap

$A \rightarrow B$ is **ONLY False** if the premise A is True but the conclusion B is False.

T $\rightarrow$ T = T
F $\rightarrow$ T = T
F $\rightarrow$ F = T
T $\rightarrow$ F = F

LOGIC

The Tableau Method

To prove formula A is valid, assume it is false ($F[A]$) and decompose. If every path closes with a contradiction (e.g., $T[p]$ and $F[p]$ appear in the same path), the original formula is **Valid**.

Core Decomposition Rules

Non-Branching (Stack)

- $T[p \wedge q] \rightarrow T[p], T[q]$
- $F[p \vee q] \rightarrow F[p], F[q]$
- $F[p \rightarrow q] \rightarrow T[p], F[q]$

Branching (Fork)

- $T[p \vee q] \rightarrow T[p] | T[q]$
- $F[p \wedge q] \rightarrow F[p] | F[q]$
- $T[p \rightarrow q] \rightarrow F[p] | T[q]$

Result Interpretation: If any branch remains open after full decomposition, the formula is Invalid. The literals on that open branch represent a counter-example.

GRAPHS

Graph Theory Fundamentals

Classification

- **Directed:** Edges have arrows indicating specific direction.
- **Undirected:** Edges are simple lines (bidirectional).
- **Weighted:** Edges have associated numerical costs.

Topology

- **Connected:** A path exists between every node pair.
- **Isolated Node:** A vertex with no edges.
- **Cyclic:** Graph contains at least one loop (cycle).

Formal Definition Example

V = {1, 2, 3, 4}
E = {(1,2), (2,3), (3,1), (4,4)}

Analysis: This graph is **Directed**, **Disconnected** (node 4 has no connection to others), and **Cyclic** (contains a triangle 1-2-3 and a self-loop at 4).

TREES

Binary Trees & Structures

The Tree Concept

"A tree is a connected, acyclic, undirected graph. For n nodes, it always contains $n - 1$ edges."

Binary Tree Constraint

Every node in a binary tree is restricted to **at most 2 children** (Left and Right).

Tree Depth: The length of the longest path from the root to any leaf node (count the number of edges).
Rooted Tree: A tree where one node is defined as the "start" or parent of all.
Unique Path Property: In any tree, there is exactly one unique path between any two nodes.

NETWORKS

Network Latency Graphs

Weighted Directed Edges

Modeling data transmission: Edges flow from **Lower Latency** origins to **Higher Latency** destinations.

> **Edge Weight (w)** = $Latency(Dest) - Latency(Source)$

Example Logic:

- Node A (Origin): 10ms
- Node B (Node): 25ms
- Node C (Destination): 60ms
- **Graph:** (A $\rightarrow$ B, w=15), (B $\rightarrow$ C, w=35).

Note: If a central server is used as the reference point, latencies are calculated as the direct difference from that hub.

MATRICES

Matrix Arithmetic & Algebra

Algebraic Solving

Solve matrix equations like standard algebra, but ensure you maintain matrix dimensions and order.

Equation: $3X - M_1 = 7X + M_2$
Rearrange: $-4X = M_1 + M_2$
Solve: $X = -\frac{1}{4}(M_1 + M_2)$

Addition Rule

Add elements in matching positions. **Requirement:** Matrices must have identical dimensions ($m \times n$).

Scalar Scaling

Multiply every individual element within the matrix by the constant scalar factor k.

MANAGEMENT

Critical Path Analysis (CPA)

Operational Rules

- **Earliest Start (ES):** The maximum of all possible completion times of incoming activities.
- **Critical Path:** The chain of activities where there is zero float (ES = Latest Start).

Example Decision

- → Incoming Path A: Dur. 12
- → Incoming Path B: Dur. 24
- → Incoming Path C: Dur. 18
- → **Node ES:** 24 (The project is bottlenecked by the longest path).

Exam Strategy: To identify the path nodes, look for the sequence where the Earliest Start and Latest Finish calculations match perfectly.

STATISTICS

Variance & Standard Deviation

Dataset: $\{x_1, x_2, \ldots, x_n\}$
1. Mean ($\bar{x}$) $\frac{\sum x}{n}$
2. Sample Variance (S^2) $\frac{\sum (x-\bar{x})^2}{n-1}$
3. Std Deviation (S) $\sqrt{S^2}$

Step-by-Step Sample:

Data: 10, 20, 30 ($n = 3$)
Mean: 20
Squared Diffs: $(10-20)^2 + (20-20)^2 + (30-20)^2 = 100 + 0 + 100 = 200$
Variance: $200/(3-1) = 100$
Std Dev: $\sqrt{100} = 10$

COMPUTING

Binary & Database Theory

Floating Point Normalization

Standardizing 32.25 to Mantissa/Exponent:

Integer: 32 = 100000
Frac: 0.25 = .01

1. Concatenate: 100000.01

2. Shift decimal to first significant digit: 1.0000001×2^{5}

3. **Mantissa:** 10000001, **Exponent:** 5

Database Normalization

Definition

"The systematic process of organizing data to minimize redundancy (duplication) and ensure data dependency (integrity) by splitting large tables into smaller, related ones."

Study Complete. Pass ICT2.